

ASSOCIATION ANALYSIS

INTRODUCTION – PART I

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INTRODUCTION

The following concepts will be introduced:

- ✓ **ASSOCIATION ANALYSIS**
- ✓ **FREQUENT ITEMSET**
- ✓ **ASSOCIATION RULE**



The **DIRECTOR OF A MINI-MARKET** calls you on the phone telling that the mini-market wants to change the **PRODUCTS POSITIONING** (their allocation on the shelves). However, the Director of the mini-market has read on a manual for managing mini-markets, that **making a decision about products positioning is a very difficult problem to be solved.**

The **DIRECTOR OF THE MINI-MARKET** knows that making a decision about product positioning to optimize the mini-market sales requires **to be informed about the customers shopping behavior**, i.e. to know **how the mini-market customers use to fill their baskets.**



You tell the Director of the mini-market that to help him you need transactions data, i.e., records that for each shopping session contain the set of products that are bought together, in other terms **you need to know which is the content of the MARKET BASKET.**



The **DIRECTOR OF THE MINI-MARKET** asks the **SALES MANAGER** to collect all the **mini-market transactions of the last three months** and to make them available to you on a cloud repository.

After some days you receive an e-mail message from the Sales Manager, the mail contains the URL of the cloud repository for accessing the transactions data of the mini-market for the last three months.



You access the repository and find the following two files

➤ **Transactions.txt**

row → transaction ID list of items ID

➤ **Products.txt**

row → item ID price name

Row ID	S Col0	Row ID	I Item	D Price	S ProductName
Row0	224 80 109 177 50 43 83 173 70 202 94 227 162 16 236 42 197 158 92 141 200 238 138 229 161 42 124 177 9 141	0	0	2.512	swiss cheese
Row1	56 95 106 186 103 170 69 198 186 211 83 24 78 198 233 49 87 188 84 117 118 118 196 161 159 98 232 143 231 20	1	1	52.442	Cherry coke
Row2	9 196 184 119 88 196 222 94 212 187 95 3 224 54 207 55 241 240 12 235 185 30 122 76 156 117 118 12 235 41 124	2	2	28.215	Bio Coke
Row3	228 9 193 127 163 117 24 34 204 163 48 74 69 230 231 166 117 225 88 225	3	3	3.992	Peppers
Row4	94 9 22 133 107 228 77 173 38 109 32 31 110 79 79 27 22	4	4	1.042	scrambled egg
Row5	13 184 209 20 229 207 32 162 3 54 163 20 17 81 19 86 19	5	5	3.9	alkopops(rum/cherry)
Row6	158 203 205 25 137 16 194 70 65 198 64 145 241 179 203	6	6	1.656	Pomegranate
Row7	167 117 187 12 235 231 128 17 84 173 87 66 36 145 33 10	7	7	4.267	strawberries
Row8	241 222 107 200 203 92 74 145 170 239 215 59 229	8	8	2.729	chachasa
Row9	12 235 41 95 79 133 132 12 235 98 121 138 65 188 123 16	9	9	0.847	Riesling(White wine)
Row10	145 66 71 207 103 144 82 77 6 191 212 192 106 117 128	10	10	1.733	cheddar
Row11	22 0 173 67 197 233 93 101 133 203 1 241 225 138 40 17	11	11	3.114	Gourds
Row12	48 128 100 92 88 13 225 4 55 229 117 231 27 178 117 58	12	12	1.495	shrimos



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You make the decision to develop an **ASSOCIATION ANALYSIS** study to **discover interesting and useful relationships hidden in the large amount of data** which has been made available by the Sales Manager of the mini-market.

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The uncovered relationships can be represented in form of **ASSOCIATION RULES** or sets or **FREQUENT ITEMS**.

{Beer} → **{swiss cheese}**

{antecedent} → **{consequent}**

	Transaction ID	swiss cheese	cherry coke	bio coke	Peppers	scrambled egg	Pomegranate	strawberries
Row 0	0	1	0	0	0	1	0	0
Row 1	1	0	1	0	0	0	0	1
Row 2	2	0	0	0	1	1	0	0
Row 3	3	0	0	0	0	0	1	0
Row 4	4	0	0	0	0	0	0	1
Row 5	5	0	0	0	0	0	0	0
Row 6	6	0	0	0	0	0	0	0

Peppers = 1 means the
Pepper item is included into the market
basket for the considered transaction
(Transaction ID)

swiss cheese = 0 means the swiss
cheese item is not included into the
market basket for the considered
transaction (Transaction ID)

Maybe a simplistic view of real market basket data

Market basket data can be represented in a binary format

- Row = Row ID = Transaction ID
- Column = Col0 = Item